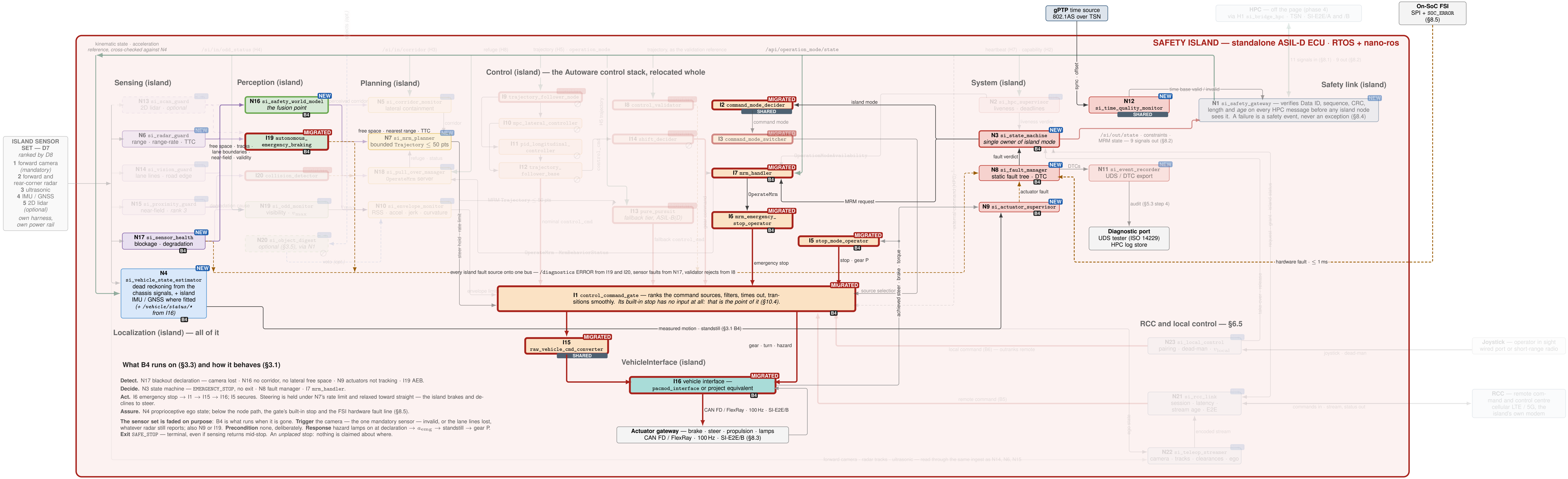
Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on — the camera — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



gPTP time source
802.1AS over TSN

HPC — off the page (phase 4)
via H1 `si_bridge_hpc` · TSN · SI-E2E/A and /B

On-SoC FSI
SPI + `SOC_ERROR`
(§8.5)

SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros

kinematic state · acceleration
reference, cross-checked against N4

`/si/in/odd_status` (H4) · `/si/in/corridor` (H3) · refuge (H8) · trajectory (H5) · `operation_mode` · trajectory, as the validation reference · `/api/operation_mode/state` · heartbeat (H7) · capability (H2) · 11 signals in (§8.1) · 9 out (§8.2)

ISLAND SENSOR SET — D7
ranked by D8
1 forward camera
(mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar
(optional)
own harness, own power rail

Sensing (island)

N13 `si_scan_guard` — 2D lidar · *optional* — NEW

N6 `si_radar_guard` — range · range-rate · TTC — NEW

N14 `si_vision_guard` — lane lines · road edge — NEW

N15 `si_proximity_guard` — near-field · *rank 3* — NEW

N17 `si_sensor_health` — blockage · degradation — NEW — B4

Perception (island)

N16 `si_safety_world_model` — *the fusion point* — NEW — B4

I19 `autonomous_emergency_braking` — MIGRATED — B4

free space · tracks · lane boundaries · near-field · validity

I20 `collision_detector` — MIGRATED

degradation cause

N19 `si_odd_monitor` — visibility · $v_{\max}$ — NEW

N20 `si_object_digest` — *optional (§3.5), via N1* — NEW

objects (opt.) · veto (opt.)

Localization (island) — all of it

N4 `si_vehicle_state_estimator` — dead reckoning from the chassis signals, + island IMU / GNSS where fitted *(+ `/vehicle/status/*` from I16)* — NEW — B4

Planning (island)

N5 `si_corridor_monitor` — lateral containment — NEW

perceived corridor · corridor

free space · nearest range · TTC

N7 `si_mrm_planner` — bounded `Trajectory` ≤ 50 pts — NEW

refuge · status

N18 `si_pull_over_manager` — `OperateMrm` server — NEW

N10 `si_envelope_monitor` — RSS · accel · jerk · curvature — NEW

$v_{\max}$ · steer hold · rate limit · envelope limit

Control (island) — the Autoware control stack, relocated whole

I9 `trajectory_follower_node` — MIGRATED

I10 `mpc_lateral_controller`

I11 `pid_longitudinal_controller`

I12 `trajectory_follower_base`

I8 `control_validator` — MIGRATED

I14 `shift_decider` — MIGRATED

I13 `pure_pursuit` — *fallback tier, ASIL-B(D)* — MIGRATED

H5 trajectory · control_cmd · gear command · MRM Trajectory ≤ 50 pts · nominal control_cmd · fallback control_cmd · `OperateMrm · MrmBehaviorStatus`

I2 `command_mode_decider` — MIGRATED — SHARED

command mode · island mode

I3 `command_mode_switcher` — MIGRATED

`OperationModeAvailability`

I7 `mrm_handler` — MIGRATED — B4

`OperateMrm` · MRM request

I6 `mrm_emergency_stop_operator` — MIGRATED — B4

emergency stop

I5 `stop_mode_operator` — MIGRATED — B4

stop · gear P

every island fault source onto one bus — `/diagnostics` ERROR from I19 and I20, sensor faults from N17, validator rejects from I8

I1 `control_command_gate` — ranks the command sources, filters, times out, transitions smoothly. *Its built-in stop has no input at all: that is the point of it (§10.4).* — MIGRATED — B4

source selection · achieved steer · brake · torque · external command (HPC) †

measured motion · standstill (§3.1 B4)

System (island)

N2 `si_hpc_supervisor` — liveness · deadlines — NEW

liveness verdict

N3 `si_state_machine` — *single owner of island mode* — NEW — B4

`/si/out/state` · constraints · MRM state — 9 signals out (§8.2)

fault verdict

N8 `si_fault_manager` — static fault tree · DTC — NEW — B4

DTCs

N11 `si_event_recorder` — UDS / DTC export — NEW

actuator fault

N9 `si_actuator_supervisor` — NEW — B4

audit (§5.3 step 4)

Diagnostic port
UDS tester (ISO 14229)
HPC log store

N12 `si_time_quality_monitor` — NEW — SHARED

sync · offset · time base valid / invalid

hardware fault · ≤ 1 ms

Safety link (island)

N1 `si_safety_gateway` — verifies Data ID, sequence, CRC, length and *age* on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4) — NEW

request · grant · island status · take-over · release

RCC and local control — §6.5

N23 `si_local_control` — pairing · dead-man · v_{local}

local command (B6) — outranks remote · joystick · dead-man

Joystick — operator in sight
wired port or short-range radio

N21 `si_rcc_link` — session · latency · stream age · E2E

remote command (B5) · commands in · stream, status out

RCC — remote command and control centre
cellular LTE / 5G, the island's own modem

encoded stream · ego state

N22 `si_teleop_streamer` — camera · tracks · clearances · ego

forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

VehicleInterface (island)

I15 `raw_vehicle_cmd_converter` — MIGRATED — SHARED

gear · turn · hazard

I16 vehicle interface — `pacmod_interface` or project equivalent — MIGRATED — B4

CAN FD / FlexRay · 100 Hz · SI-E2E/B

Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)

What B4 runs on (§3.3) and how it behaves (§3.1)

Detect. N17 blackout declaration — camera lost · N16 no corridor, no lateral free space · N9 actuators not tracking · I19 AEB.

Decide. N3 state machine — `EMERGENCY_STOP`, no exit · N8 fault manager · I7 `mrm_handler`.

Act. I6 emergency stop → I1 → I15 → I16; I5 secures. Steering is held under N7's rate limit and relaxed toward straight — the island brakes and declines to steer.

Assure. N4 proprioceptive ego state; below the node path, the gate's built-in stop and the FSI hardware fault line (§8.5).

The sensor set is faded on purpose: B4 is what runs when it is gone. **Trigger** the camera — the one mandatory sensor — invalid, or the lane lines lost, whatever radar still reports; also N9 or I19. **Precondition** none, deliberately. **Response** hazard lamps on at declaration → a_{emg} → standstill → gear P. **Exit** `SAFE_STOP` — terminal, even if sensing returns mid-stop. An *unplaced* stop: nothing is claimed about where.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated

NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B4

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘ **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

→ HPC-sourced, verified by N1
→ island sensor data
→ control command into the gate
⇢ fault · `/diagnostics`
→ actuator feedback
→ state · mode · decision
⇢ optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B4, blind emergency stop

🟥 **full colour** — named for B4 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

▨ **half-tone** — on B4's path by §3.1, but not named in §3.3

☐ **faded** — not used by B4. An edge is drawn at the weaker of its two ends. Only B4's own tag is drawn on each box.